Introduction

Artificial Intelligence · Computer Science · Sofia University

Bogomil Stoyanov

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Outline

1. Defining AI

2. A very short history

3. The course ahead

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Defining AI

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Defining AI

	like humans	**rationally**
Thinking	Model the mechanism. *Cognitive science*: does the program reason the way people demonstrably do?	Model correct inference. *Logic*: does the program derive only valid conclusions?
Acting	Model the behaviour. *Turing test*: is the program's output indistinguishable from a person's?	Model good decisions. *Agents*: does the program do the right thing, given what it knows?

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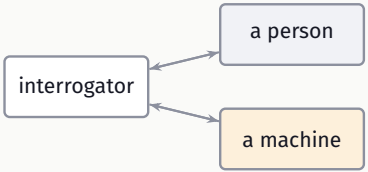
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We take the bottom-right box. It is the only one of the four that is *measurable*. We can state a goal, run the program and check whether it achieved it. "Thinks like a human" is not something you can put in a test suite.

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Acting humanly: the Turing test

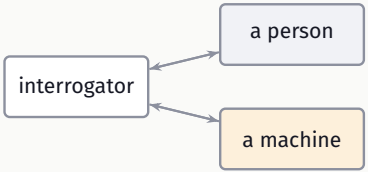


text only
which is which?

Turing, 1950: stop arguing about whether machines can think and ask whether one can be *told apart* from a person in conversation.

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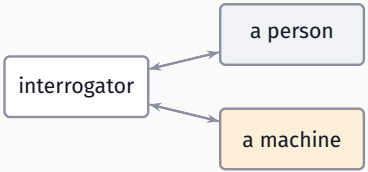
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The systems that pass it convincingly are optimised for sounding human, not for being right.

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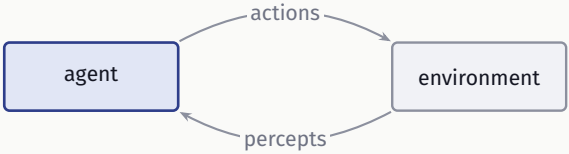
So: an agent

Agent

An **agent** perceives its environment through **sensors** and acts on it through **actuators**. Its behaviour is a function

$$f : \mathcal{P}^* \longrightarrow \mathcal{A}$$

from every possible *history* of percepts to an action.



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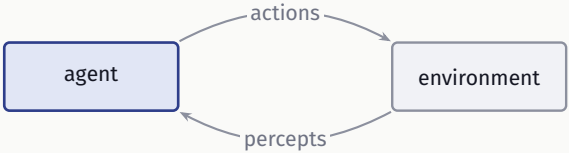
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Rational does not mean omniscient, clairvoyant or successful. It means: take the action expected to do best *given the percepts so far and the knowledge available*.

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PEAS: how to pin down a task before coding it

An automated taxi

Performance: safety, reaching the destination, legality, fuel, passenger comfort.

Environment: roads, other traffic, pedestrians, weather.

Actuators: steering, accelerator, brake, indicators, horn.

Sensors: cameras, lidar, speedometer, GPS, engine sensors.

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The second one fits on a slide. That is the whole reason the course starts with puzzles and games: the environment is small enough to be *written down exactly*, so we can study the algorithm instead of the mess.

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Environment types

	Crossword	Chess	Poker	Taxi
Fully observable?	Yes	Yes	No	No

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Read the columns left to right: that is the syllabus. Sessions 02–04 live in the first column, session 06 adds the opponent, and the last column is why machine learning exists.

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A very short history

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Two AI winters

1943	McCulloch & Pitts: a Boolean circuit model of a neuron
1950	Turing's "Computing Machinery and Intelligence"
1956	The Dartmouth workshop: the term *artificial intelligence*
1952–69	Early optimism: theorem provers, game players, general problem solvers
1966–73	**Reality:** toy problems do not scale; NP-hardness; funding withdrawn
1969–79	Knowledge-based systems: put the domain in, not just the search
1980–88	Expert systems become an industry
1988–93	**The AI winter:** brittle systems, unmaintainable rule bases
1988–	Probability and decision theory; agents; statistical learning
2012–	Deep learning: the data and the compute finally arrive

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Both busts had the same cause: a method that worked on a small example was assumed to scale. Session 02 will make that concrete: a search that is fine at depth 6 needs a petabyte of memory at depth 12.

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The course ahead

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The twelve sessions

Search: find a sequence of actions

02 uninformed → 03 informed → 04 CSP → 05 genetic → 06 games

Learning: find the function itself

07 intro to ML → 08 k-NN → 09 naive Bayes → 10 decision trees → 11 clustering

12 neural networks

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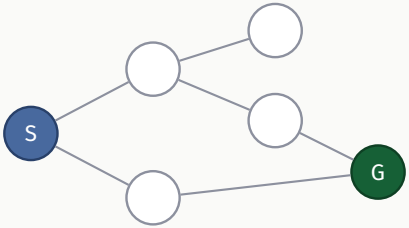
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The split is the one from the table two slides ago. In the top row *we* supply the knowledge and the program searches. In the bottom row the program is handed data and has to supply the knowledge itself.

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The one idea the next five sessions rest on



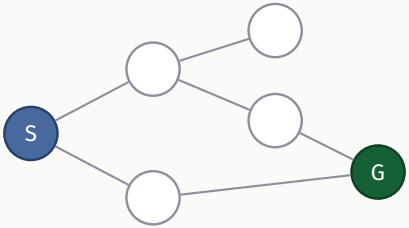
which sequence of actions gets from S to G?

Almost every task in the top row of that chart can be forced into the same shape:

- a set of **states** the world can be in,
- **actions** that move between them,
- a test for which states count as **done**.

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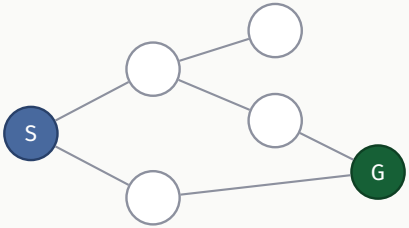
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That object is a search problem, and session 02 is entirely about it.

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